

MEDICAL DEVICE MARKETING
UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

2. Financial Targets

Financial targets are the **specific financial goals established for a product or project**.

They help determine whether the product is commercially viable.

Important financial targets

1. Development budget

Maximum amount allocated for product development.

2. Manufacturing cost

Cost required to manufacture one unit.

3. Selling price

Price at which the product is sold.

4. Revenue target

Expected income from product sales.

5. Profit target

Expected profit from the product.

6. Break-even point

The sales level at which total revenue equals total cost.

7. Return on Investment (ROI)

Measures the return generated from an investment.

$$\text{ROI} = (\text{Net Return} / \text{Investment}) \times 100$$

Example

Suppose:

- Manufacturing cost = ₹8,000/device
- Selling price = ₹15,000/device
- Expected sales = 1,000 devices

The company can estimate its revenue, contribution margin, operating costs, and expected profit before deciding whether to proceed with the project.

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3. Physician Input

Introduction

Physician input refers to obtaining information, suggestions, feedback, and clinical requirements from doctors during medical product development.

Physicians are important stakeholders because they understand the **clinical problem, patient requirements, workflow, treatment procedures, and limitations of existing technologies**.

Importance of Physician Input

1. Identification of clinical problems

Physicians can identify problems that require technological solutions.

Example:

A physician may identify that existing patient monitoring systems provide information too late for early intervention.

2. Defining product requirements

Doctors can specify:

- Required parameters
- Accuracy
- Response time
- Clinical functionality
- Patient population
- Desired alerts

3. Product design

Physician feedback helps engineers develop a device suitable for actual clinical environments.

4. Usability

Doctors can evaluate:

- User interface
- Display
- Controls
- Workflow
- Ease of operation

5. Clinical validation

Physicians may participate in clinical evaluation and provide feedback regarding clinical performance.

6. Risk identification

Doctors can identify possible clinical hazards and inappropriate uses.

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7. Product acceptance

A product developed with appropriate clinical input is more likely to be accepted by healthcare professionals.

Methods of Obtaining Physician Input

- Interviews
- Questionnaires
- Focus groups
- Clinical observations
- Product demonstrations
- Advisory panels
- Prototype evaluation
- Clinical studies

Example

For a **smart prosthetic limb**, engineers may develop the sensors and control system, while physicians and rehabilitation professionals can provide input regarding:

- Patient comfort
- Gait requirements
- Safety
- Socket fit
- Rehabilitation needs
- Clinical usability

4. FDA Approval

Introduction

The **U.S. Food and Drug Administration (FDA)** regulates medical devices marketed in the United States. FDA's Center for Devices and Radiological Health (CDRH) regulates firms that manufacture, repack, relabel, or import medical devices for the U.S. market.

An important point for students is that **not every medical device receives an "FDA approval."** Depending on the device and regulatory pathway, the product may be **510(k) cleared, De Novo classified, PMA approved, or exempt from a premarket submission.**

Objectives of FDA Regulation

FDA regulation aims to ensure that medical devices:

- Are reasonably safe
- Are effective for their intended use

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- Mechanical testing
- Electrical safety testing
- Biocompatibility
- Software verification
- Electromagnetic compatibility
- Sterilization validation
- Packaging testing
- Performance testing
- Animal testing where appropriate

The exact testing depends on the device.

Step 5 – Clinical Evaluation/Investigation

For devices requiring clinical evidence, clinical investigations may be conducted to evaluate:

- Safety
- Effectiveness
- Performance
- Patient outcomes
- Adverse events

Where applicable, an **Investigational Device Exemption (IDE)** permits certain investigational devices to be used in clinical studies in accordance with FDA requirements.

Step 6 – Quality Management System

Manufacturers must establish an appropriate quality system.

A major current development is that FDA's **Quality Management System Regulation (QMSR)** became effective on **February 2, 2026**. QMSR amended 21 CFR Part 820 and incorporates by reference **ISO 13485:2016** for medical device quality management systems.

Therefore, students should now understand:

21 CFR Part 820 → QMSR → incorporates ISO 13485:2016

The QMS supports areas such as:

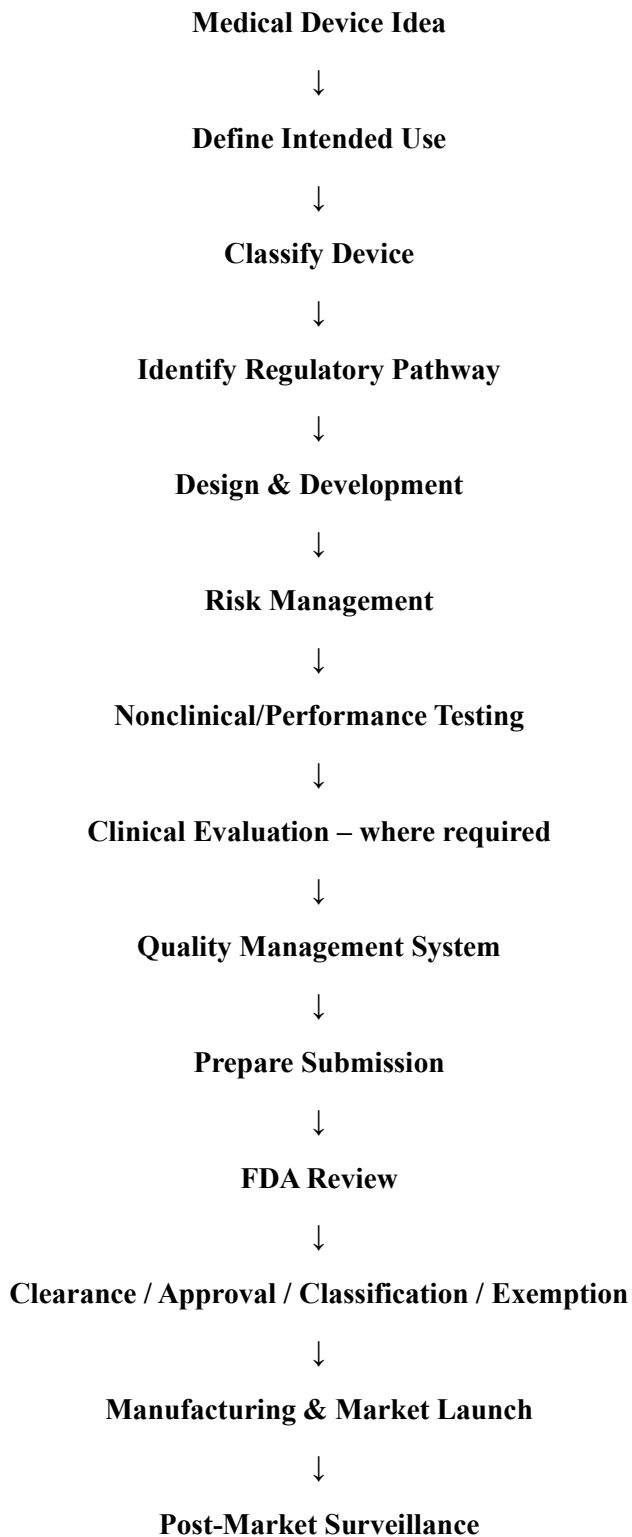
- Design and development
- Manufacturing
- Quality control
- Documentation

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- Monitoring of device safety and performance

FDA states that it is responsible for assuring that medical devices available in the United States remain safe and effective throughout their total product lifecycle.

Simplified FDA Pathway



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Important Difference: Clearance vs Approval

Term	Common pathway	Meaning
FDA Clearance	510(k)	Device has been determined substantially equivalent to an appropriate predicate
FDA Approval	PMA	FDA has approved the PMA based on applicable evidence of safety and effectiveness
De Novo Classification	De Novo	New device type classified through the De Novo pathway
Exempt	Certain devices	No applicable premarket submission, subject to the conditions and limitations of the exemption

5. Reimbursement

Introduction

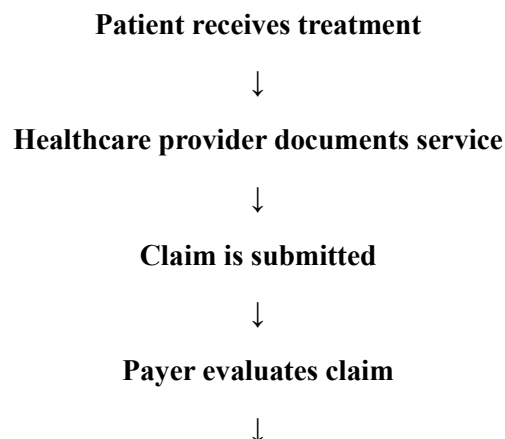
Reimbursement refers to the payment made to healthcare providers or organizations for eligible healthcare services, procedures, or products.

For a medical device company, reimbursement is important because a product may be clinically effective but commercially unsuccessful if customers cannot obtain adequate financial coverage for its use.

Major Participants

- Patient
- Doctor
- Hospital
- Insurance company/payer
- Government
- Medical device manufacturer

Basic Process



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Claim approved/rejected



Payment made

Importance of Reimbursement

Reimbursement influences:

- Patient affordability
- Hospital purchasing
- Product adoption
- Market size
- Revenue
- Product pricing

Example

A new cardiac monitoring device may provide excellent clinical benefits. However, if hospitals or patients cannot recover the cost through an applicable payment mechanism, adoption may remain low.

6. Managing Issues

During product development and commercialization, several issues can arise.

Common issues

- Technical failures
- Component shortages
- Design changes
- Budget overruns
- Regulatory delays
- Testing failures
- Customer complaints
- Manufacturing problems
- Software errors
- Supplier delays

Issue Management Process

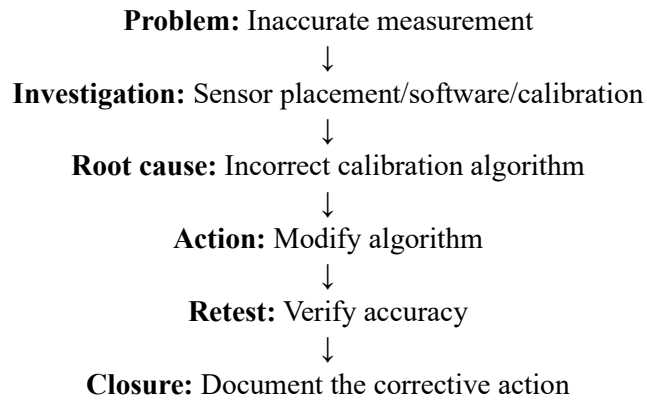
Identify issue → Analyze cause → Determine impact → Prioritize → Develop corrective action → Assign responsibility → Implement action → Monitor → Close issue

Example

If a prototype shows inaccurate SpO₂ readings:

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7. Meeting Deadlines

Meeting deadlines means completing project activities within the planned time.

Methods

1. Develop a project schedule

Define:

- Activities
- Start date
- End date
- Dependencies
- Responsible person

2. Prioritize critical activities

Identify activities that directly affect the final delivery date.

3. Monitor progress

Use:

- Gantt charts
- Milestone tracking
- Weekly meetings
- Progress reports

4. Allocate resources properly

Ensure availability of:

- Personnel
- Equipment
- Materials

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- Budget

5. Manage risks

Identify potential delays before they occur.

6. Take corrective action

If an activity is delayed:

- Reallocate resources
- Change sequence
- Add manpower
- Revise schedule
- Address technical problems

Example

If clinical testing is delayed by two weeks, the team should assess whether documentation, software development, or manufacturing activities can proceed in parallel without compromising quality.

8. Product Launch

Introduction

Product launch is the process of introducing a new product into the target market and making it available to customers.

For a medical device, product launch is more than simply selling the product. It involves:

- Regulatory readiness
- Manufacturing readiness
- Market preparation
- Pricing
- Distribution
- Sales training
- Customer support
- Product communication
- Post-market monitoring

Stages of Product Launch

1. Market Preparation

Before launching the product, the company should understand:

- Target customers

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- Market size
- Competitors
- Customer requirements
- Pricing
- Distribution channels
- Market barriers

2. Regulatory Readiness

The company must ensure that applicable regulatory requirements have been satisfied before marketing the device in the relevant jurisdiction.

This includes appropriate:

- Regulatory status
- Labeling
- Documentation
- Quality systems
- Manufacturing controls

3. Manufacturing Readiness

The company should ensure that sufficient quantities can be produced with consistent quality.

Activities include:

- Supplier qualification
- Production planning
- Quality inspection
- Packaging
- Sterilization where required
- Inventory planning

4. Pricing Strategy

The company determines the price based on:

- Manufacturing cost
- Competitor prices
- Customer willingness to pay
- Clinical value

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- Distribution cost
- Market positioning
- Reimbursement environment

5. Sales Preparation

The sales team should understand:

- Product features
- Clinical benefits
- Product limitations
- Target customers
- Competitors
- Pricing
- Demonstration procedure
- Customer objections

6. Marketing and Promotion

Promotion may include:

- Product demonstrations
- Conferences
- Medical exhibitions
- Digital marketing
- Scientific publications
- Healthcare professional education
- Sales presentations

Medical device claims should remain consistent with applicable regulatory requirements and authorized/intended uses.

7. Distribution

The company determines how the product will reach customers.

Possible channels include:

Manufacturer → Distributor → Hospital → Patient

or

Manufacturer → Hospital

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or

Manufacturer → Clinic → Patient

8. Product Training

Healthcare professionals may need training on:

- Installation
- Operation
- Maintenance
- Safety
- Troubleshooting
- Clinical use

9. Launch

The product is officially introduced to the target market.

Launch activities may include:

- Product announcement
- Demonstrations
- Distributor activation
- Sales campaigns
- Hospital visits
- Customer training

10. Post-Launch Monitoring

After launch, the company monitors:

- Sales
- Customer feedback
- Complaints
- Device failures
- Returns
- Adverse events
- Market acceptance
- Profitability

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Product Launch Framework



Factors Determining Successful Product Launch

Technical readiness

The device should perform according to its specifications.

Clinical readiness

Healthcare professionals should understand its clinical value.

Regulatory readiness

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Required regulatory requirements must be satisfied.

Financial readiness

The company should have adequate financial resources.

Manufacturing readiness

Production should meet expected demand.

Sales readiness

Sales personnel should be properly trained.

Customer support

Installation, training, maintenance, and troubleshooting should be available.

Market readiness

There should be sufficient customer demand.

9. Forecasting

Forecasting is the process of **estimating future demand, sales, revenue, or market requirements**.

It helps companies determine how many products they should manufacture and how much inventory they should maintain.

Types of Forecasting

1. Qualitative forecasting

Based on expert opinion and market knowledge.

Methods include:

- Expert opinion
- Sales-force opinion
- Customer surveys
- Delphi method

2. Quantitative forecasting

Uses historical data and mathematical/statistical techniques.

Examples:

- Moving averages
- Trend analysis
- Regression analysis
- Time-series analysis

Factors Affecting Medical Device Forecasting

- Number of hospitals

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- Patient population
- Disease prevalence
- Competitor products
- Product price
- Healthcare policies
- Reimbursement
- Technology adoption
- Economic conditions

Example

If a company sold:

- 500 units in Year 1
- 700 units in Year 2
- 900 units in Year 3

The company may use historical trends and market growth assumptions to estimate future demand.

Importance

Forecasting helps with:

- Production planning
- Inventory control
- Manpower planning
- Financial planning
- Sales targets
- Supply-chain management

10. Pricing

Pricing is the process of determining the **amount a customer must pay for a product or service.**

Objectives of Pricing

- Recover costs
- Generate profit
- Gain market share
- Compete effectively
- Reflect product value
- Encourage adoption

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Factors Affecting Pricing

1. Manufacturing cost

Higher production costs generally increase the minimum viable selling price.

2. Research and development cost

Medical device companies must recover development investment over time.

3. Competitor pricing

The company should understand prices of competing products.

4. Customer willingness to pay

Customers may pay more for significant clinical and economic benefits.

5. Clinical value

A product that significantly improves outcomes may justify a higher price.

6. Reimbursement

The available payment mechanism can strongly influence practical affordability and adoption.

7. Distribution cost

Distributor margins, transportation, installation, and service affect the final price.

Common Pricing Approaches

Cost-plus pricing:

Cost + desired profit margin

Competition-based pricing:

Price based on competitors

Value-based pricing:

Price based on perceived/ demonstrated customer and clinical value

Penetration pricing:

Lower initial price to encourage market adoption

Premium pricing:

Higher price for differentiated products with strong value or positioning

11. Product and Sales Support

Introduction

Product and sales support refers to the technical, clinical, marketing, training, and service assistance provided to customers and sales teams before and after a product is sold.

Product Support

Product support includes:

- Installation

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- User training
- Operation manuals
- Technical assistance
- Preventive maintenance
- Corrective maintenance
- Spare parts
- Software updates
- Troubleshooting
- Warranty service

Sales Support

Sales support helps sales personnel successfully communicate and demonstrate the product.

It includes:

- Product brochures
- Technical specifications
- Demonstration units
- Product presentations
- Training for sales personnel
- Competitor comparison
- Frequently asked questions
- Clinical evidence
- Customer case studies
- Pricing information

Importance

1. Improves customer confidence

Customers feel more confident when technical support is available.

2. Increases sales effectiveness

Sales personnel can answer technical and clinical questions.

3. Reduces product misuse

Proper training helps users operate devices correctly.

4. Improves customer satisfaction

Fast service and troubleshooting improve the customer experience.

5. Encourages repeat purchases

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Satisfied customers are more likely to purchase additional products.

Example

For an ultrasound system, sales support may include product demonstration and clinical application training, while product support includes installation, preventive maintenance, troubleshooting, and service.

12. Managing an Existing Product Line

Introduction

A **product line** is a group of related products marketed by the same company.

For example, a medical technology company may have a product line containing:

- Basic patient monitor
- Advanced patient monitor
- ICU monitor
- Wireless monitor

Managing an existing product line means continuously monitoring and improving the products to maintain their **market position, profitability, quality, and customer satisfaction**.

Major Activities

1. Monitor sales performance

Track:

- Sales volume
- Revenue
- Profit
- Market share
- Growth rate

2. Monitor customer feedback

Collect information about:

- Product satisfaction
- Complaints
- Usability
- Reliability
- Desired improvements

3. Product improvement

Improve:

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- Hardware
- Software
- User interface
- Accuracy
- Connectivity
- Reliability

4. Manage product variants

Companies may offer different versions for different customer segments.

5. Manage pricing

Prices may need adjustment based on:

- Competition
- Cost
- Demand
- Market position

6. Inventory management

Maintain sufficient stock without creating excessive inventory.

7. Competitor monitoring

Track:

- New competitor products
- Technology changes
- Pricing
- Features
- Market strategy

8. Product life-cycle management

A product generally passes through:

Introduction → Growth → Maturity → Decline

At each stage, the company may modify its marketing and product strategy.

9. Decide whether to modify or discontinue

If a product has declining demand, high maintenance costs, or is replaced by better technology, the company may:

- Improve it
- Reposition it
- Reduce its price

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- Replace it
- Discontinue it